Alignment Beyond Extraction: From Preference Optimization to Situated Value Elicitation

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Abstract

The dominant alignment paradigm treats human values as stable entities that can be extracted through preference signals and encoded into AI systems. I argue that this foundationalist epistemology systematically misrepresents the phenomenon it claims to measure. Drawing on philosophy of science, HCI methodology, and applied experience in AI model evaluation, I propose alignment as *situated reflective elicitation*: an ongoing, co-evolutionary process in which values are understood as patterns that emerge through reflection on action in context. I outline why the preference-optimization pipeline is epistemologically brittle, why the choice of whose values to align with is itself a consequential value judgment, and how qualitative methods deployed at scale can produce richer alignment signals and concrete paths to operationalization through contextual post-training and continual learning. I close by tracing how alignment transforms as AI capabilities advance from conversational assistance through autonomous digital and physical agency to societal coordination, and why each transition demands a different epistemological toolkit.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

human-AI alignment, value elicitation, qualitative methods, epistemology, co-evolution

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1 Introduction

Research on human-AI alignment has made significant technical progress, yet it rests on epistemological assumptions that deserve more scrutiny than they typically receive. This paper engages with two of the BiAlign workshop's core themes: *Value-Centered Alignment Objectives*, asking which human values should guide alignment and how they can be articulated [16], and *Dynamic Co-Evolution of Human-AI Futures*, asking how alignment goals and practices must evolve as humans and AI systems mutually adapt.

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The methods at the center of current alignment practice share a common structure: elicit human preferences through discrete comparison tasks, aggregate these preferences into a reward signal or set of principles, and optimize model behavior accordingly [2, 13, 14]. The implicit epistemology is foundationalist. Values are treated as pre-existing objects that can be surfaced, measured, and encoded into systems through optimization.

Shen et al.'s bidirectional alignment framework [16] rightly identifies that alignment must operate in two directions: shaping AI to reflect human values, and supporting humans in understanding and critically engaging with AI systems. This corrects unidirectional framings. However, even a bidirectional paradigm can inherit foundationalist assumptions about what values *are* and how they can be *known*. If alignment in both directions relies on the same epistemological infrastructure, namely preference rankings, constitutional rules, and annotator judgments, it reproduces the same blind spots in stereo.

This paper argues that the field needs a different *epistemology* of alignment. I propose what I call *situated reflective elicitation*: an approach that treats values as contextually emergent, paradigm-dependent, and co-constructed through interaction. I ground this argument in applied experience conducting AI model evaluations at scale [11], and in the philosophical observation that the choice of *whose* values to align with, and at what level of aggregation, is itself a value judgment that current technical approaches render invisible.

2 The Epistemological Problem with Current Alignment

2.1 The Preference-Optimization Pipeline

To make the epistemological critique precise, it helps to spell out what current alignment methods actually do. In RLHF [13], human annotators compare pairs of model outputs and indicate which they prefer. These pairwise preferences train a reward model that assigns scalar scores, and the language model is fine-tuned via reinforcement learning to maximize those scores. DPO [14] collapses this into a single step by directly optimizing the language model on preference data, treating the model itself as an implicit reward function. Constitutional AI [2] replaces some human annotation with model self-critique guided by written principles, then applies RLHF on the revised outputs.

These methods differ in important ways. DPO eliminates the explicit reward model. Constitutional AI introduces principles that go beyond raw preference data. Yet all three share a core epistemological assumption: that human values can be adequately captured through preference orderings over discrete text outputs, and that these orderings generalize across contexts. The pairwise comparison format requires that values be expressible as relative rankings. The reward model (or its implicit equivalent) requires that these rankings compress into a scalar signal. The optimization objective requires that this signal be stable enough to serve as a fixed target.

2.2 Why the Assumption Fails

To be clear, these methods have produced real gains. RLHF-trained models are measurably more helpful and less harmful than their base counterparts, and the practical value of preference optimization for filtering egregious outputs is well established. The epistemological concern is about the *ceiling* of this approach, not its floor. Reward models exhibit well-documented brittleness: reward hacking, where models find high-reward behaviors that violate the intent of the preference signal, and catastrophic Goodhart effects, where policies achieve arbitrarily high reward while producing no additional utility [5]. Sycophancy, where models learn to agree with users rather than provide accurate information, has been linked to the same optimization dynamics [15]. These failures are symptoms of a representational mismatch between the complexity of human values and the signal used to approximate them.

Empirical work on contextual value alignment reinforces the concern. Shen et al.'s ValueCompass framework measured alignment between humans and large language models across countries, scenarios, and value dimensions grounded in Schwartz's theory of basic values. The results showed moderate alignment at best (highest F1 = 0.529), with substantial variation across national contexts and use cases [17]. Values, in practice, are not stable across contexts. Alignment measured in one setting does not predict alignment in another.

In my own work evaluating AI models across computer vision, media generation, and conversational AI, I have consistently observed that expert annotator judgments diverge from end-user evaluations in structured and informative ways [11]. In one case, traditional annotation-based evaluation of a conversational AI model indicated strong wins against competitors across use cases, while evaluations conducted with actual product users showed ties or losses in the same categories. The annotators were measuring stated criteria faithfully. The users were responding to experienced quality. These are different phenomena. The gap between them is epistemological: preference elicitation under artificial conditions captures a different kind of signal than situated evaluation in context, and no amount of annotator training resolves a mismatch at that level.

2.3 The Aggregation Problem as a Value Judgment

Current alignment approaches implicitly choose a level of aggregation at which to define "human values": individual preferences, population-level distributions, expert-authored principles, or civilizational norms. They then proceed as though this choice were neutral. It is not. The level of aggregation determines what alignment *means*, and different choices produce different, sometimes contradictory, alignment targets.

Sorensen et al. demonstrate empirically that standard alignment procedures reduce distributional pluralism in model outputs [18]. Optimizing for aggregate preferences compresses the space of values a model can represent, effectively silencing minority perspectives in the name of alignment.

The aggregation choice is consequential beyond fairness concerns. It determines whether alignment prioritizes individual autonomy or collective welfare, whether cultural specificity is preserved or averaged away, and whether the values of system builders are privileged over those of users. These are political and ethical decisions. The foundationalist framing obscures them by treating the preference signal as a transparent window onto human values when it is a constructed artifact shaped by its designers' choices. Situated elicitation, as I propose in Section 3, does not resolve the aggregation problem. It makes the question more *visible*, which is itself a contribution: forcing alignment practitioners to confront choices that the current paradigm allows them to elide.

2.4 Stated Preferences and Operative Values

A further limitation is that what people *say* they value and what actually *guides their behavior* are often different [7]. Preference elicitation under artificial conditions captures stated preferences, which may reflect social desirability or simplified heuristics. The operative values that shape real-world interaction are often invisible to the person holding them until a situation makes those values salient. User research methodologies emphasize ecological validity for exactly this reason: studying behavior in naturalistic contexts, supplementing self-report with observation of actual use [12]. Alignment research has largely bypassed this lesson, building its measurement infrastructure on the kind of decontextualized elicitation that HCI has spent decades learning to move beyond.

3 Alignment as Situated Reflective Elicitation

If values are not stable objects waiting to be extracted, what are they, and how should alignment engage with them? I propose *situated reflective elicitation*: a process in which values are surfaced through structured reflection on actual interactions and the principles guiding them.

3.1 Values as Contextually Emergent

The philosophical ground for this proposal draws on anti-foundationalist epistemology and the Kuhnian insight that observation is paradigm-dependent [9]. What counts as "aligned" behavior is not fixed. It depends on the context of interaction, the expectations shaped by prior experience, and the conceptual frameworks through which users interpret AI behavior. A response that feels aligned in a casual conversational context may feel invasive in a medical one. A model that preserves user autonomy in information-seeking tasks may feel unhelpful when the user needs decisive guidance.

This does not make alignment arbitrary. It makes alignment *relational*: a property of the fit between a system and the specific human context it operates in. Evaluating alignment requires understanding the context richly enough to assess fit, and discrete preference signals, by design, cannot provide this.

3.2 From Extraction to Reflection

If values are contextually emergent, alignment requires a process that engages with them on their own terms: facilitating reflection on past actions and the principles guiding them, surfacing the *shape* of values in context across diverse user populations. The goal is a textured understanding of how different communities experience AI systems, what they expect, what they reject, and why. Several recent efforts point in this direction. The STELA framework demonstrates community-centered norm elicitation through participatory processes with historically underrepresented groups [3]. Ma et al. propose human-AI deliberation as an alternative to static preference elicitation, using structured dialogue to surface and refine values iteratively [10].

These efforts share terrain with the growing literature on pluralistic alignment [18], which seeks to make the preference-optimization pipeline more representative by incorporating diverse voices and preserving distributional variation in model outputs. The contribution of situated reflective elicitation is at a different level. Pluralistic approaches diversify the *inputs* to the existing epistemological infrastructure: more populations, more contexts, less aggressive aggregation. Situated elicitation changes the *mode* of engagement, from discrete preference signals to structured reflection that generates evaluative dimensions not known in advance. Both are valuable, and they are complementary rather than competing. But diversifying inputs to a pipeline that still compresses values into preference orderings does not resolve the epistemological limitations described in Section 2.

In my applied work, I have seen this kind of approach produce materially different outcomes than preference-based evaluation. When developing evaluation frameworks for audiovisual AI models, qualitative interviews with domain experts and end users surfaced quality criteria, including the distinction between functional, experiential, and emotional dimensions of interaction, that no preference ranking would have revealed, because the dimensions themselves were not known in advance. The evaluation framework emerged from reflection on what experts and users experienced and valued. It subsequently guided both quantitative evaluation and model development decisions [11]. This is alignment in practice: developing a grounded understanding of what matters and letting that understanding shape the system iteratively.

3.3 Operationalizing Richer Signals

A common objection is that qualitative methods cannot be translated into model behaviors at scale. The challenge is real, but it is the same translation problem that product teams, policy makers, and institutional designers navigate routinely when moving from nuanced understanding to systematic action. Several paths are available.

First, the signal produced by situated elicitation can inform *contextual post-training*. Rather than optimizing a model toward a single aggregate preference target, post-training could embed values alongside the contexts in which those values apply. This requires models that maintain awareness of context at inference time and can hold genuinely conflicting values, applying each in the situations where relevant users find them appropriate. The architecture for this is not speculative. Context-conditioned generation and mixture-of-experts approaches already demonstrate that models can maintain and selectively deploy different behavioral profiles depending on inputs. Extending this to value representation is a research challenge, but a tractable one.

Second, alignment can be understood as a special case of *continual learning* [8]. If models are designed to learn from ongoing interaction, including structured reflection with users on the fitness of actions taken, alignment becomes a continuous update process rather than a one-time optimization. A new user's values could be elicited through an initial reflective process, then refined as the system observes decisions in context and solicits posterior updates on their appropriateness. This framing connects alignment to an active area of ML research and suggests that progress on continual learning can drive progress on alignment, and that grounding continual learning in value-relevant signals ensures those advances serve broader human needs.

Third, qualitative-at-scale methods [6], enabled by recent advances in AI-assisted qualitative analysis, can produce structured value maps across populations. These maps can directly inform how loss functions are constructed, what evaluation criteria are prioritized, and what deployment contexts require additional safeguards. The translation from understanding to behavior is indirect. It operates at the level of structured constraints on training and deployment decisions, the same kind of translation that institutions make when moving from research findings to policy.

4 Co-Evolution Across Capability Horizons

The epistemological problem I have described is compounded by the fact that alignment is not a fixed target. As AI capabilities advance, the nature of the alignment challenge transforms qualitatively. I trace this trajectory briefly here, not to develop each stage fully, but to show that the epistemological argument of Sections 2–3 has different and escalating implications at each capability horizon.

This transformation is already visible in recent history. Early chatbots faced an alignment challenge that amounted to *do not be obviously wrong*. As models improved, the challenge shifted to *do not cause obvious harm*. Current frontier models face something more subtle: *act consistently with a broadly shared moral framework unless the user's context demands otherwise*, reflecting a default ethical consensus while remaining responsive to legitimate variation in user needs and values.

Each of these transitions changed what "alignment" referred to. The next transitions will be more dramatic. As models gain the ability to take extended digital action, automate multi-step workflows, and eventually operate physically in the world, the alignment challenge becomes: *understand and advance the user's aims in a world where the equilibrium is more dynamic than the user can fully track*. Alignment may require a user to trust that an unintuitive output is actually what they would want given information they do not yet have. It may require a model that truly understands a user's values to act toward the collective good over that user's immediate preference, and to be able to justify why.

This trajectory has implications for the epistemological argument of this paper. Preference optimization was developed for, and is roughly adequate to, the "do not cause obvious harm" stage. It struggles already at the "act within a broadly shared moral framework" stage, as evidenced by the brittleness and context-sensitivity documented in Section 2. For the agentic and coordinative stages ahead, it is categorically insufficient. Aligning systems that act, persist, and adapt over time requires an alignment epistemology that can itself persist and adapt [4].

A further consideration: as personalized AI systems amplify each individual's capacity to enact their values, the resulting dynamics may alter the equilibrium of whose values shape collective outcomes. Per-individual AI that faithfully enacts each person's values could create a fundamentally new landscape for collaboration, contestation, and governance, implying another layer of alignment entirely: resolving competing values when many individually-aligned agents interact at scale [1].

5 Toward a Research Agenda

The epistemological reorientation I have described implies several concrete research directions.

Longitudinal value elicitation. If values co-evolve with AI capabilities, alignment research needs longitudinal designs that track how users' expectations, trust, and evaluative criteria shift through sustained interaction. Cross-sectional preference snapshots are insufficient for a co-evolutionary process.

Making aggregation choices explicit and contestable. Alignment processes should surface the choice of whose values they optimize for, justify it, and provide mechanisms for affected parties to contest it. Participatory and deliberative methods from democratic theory offer relevant models [1, 10].

Contextual post-training and continual alignment. The connection between situated value elicitation and continual learning should be pursued as a concrete research program, producing models that learn from structured reflection with users and maintain conflicting values alongside the contexts in which each applies.

Situated evaluation paradigms. Alignment should be assessed as a dynamic relationship, requiring evaluation methods sensitive to context, population, and time, closer to how HCI evaluates usability across diverse user groups than how ML evaluates model performance on static benchmarks.

6 Conclusion

Alignment is a problem of epistemological fit between how we conceptualize values and how values actually operate in situated human-AI interaction. The foundationalist paradigm of extraction, encoding, and optimization has enabled real progress. Its limitations grow more consequential as AI systems grow more capable, as the distance between the simplicity of the preference signal and the complexity of the alignment challenge widens at every step of the capability trajectory. A situated epistemology of alignment, grounded in reflective elicitation and operationalized through contextual post-training and continual learning, offers a path toward alignment that is responsive to context, respectful of plurality, and capable of evolving alongside the systems it governs. The urgency is to align with greater fidelity to the complexity of what we are aligning with, before the gap between our epistemological tools and our systems' capabilities becomes too wide to bridge.

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